

Nikhil Kumar Thota

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Education

Ph.D. Chemical and Biomolecular Engineering <i>Johns Hopkins University, Baltimore, MD</i>	Dec 2025
M.S.E Chemical and Biomolecular Engineering <i>Johns Hopkins University, Baltimore, MD</i>	Dec 2021
B.Tech. Chemical Engineering <i>Manipal Institute of Technology, Manipal, India</i>	May 2019

Research Experience

Ph.D. Candidate <i>Advisors: Dr. Rigoberto Hernandez (Johns Hopkins), Dr. Paulette Clancy (Johns Hopkins)</i> - Developed nested autoencoders (NestedAE) and nested heteroencoders (NestedHE) to construct low dimensional material embeddings for materials discovery from high dimensional and multi-scale datasets. - Developed a multi-agent reinforcement learning (MARL) algorithm to construct sparse feature subsets from high dimensional datasets.	Jan 2022 – Dec 2025
Ph.D. Intern <i>Advisor: Dr. Jinhui Tao (PNNL, Richland), Dr. Rigoberto Hernandez (Johns Hopkins)</i> Trained bayesian neural networks to predict binding free energies of polypeptoid sequences to calcite surfaces.	Jul 2024 – Sep 2024
M.S.E Student <i>Advisor: Dr. Rigoberto Hernandez (Johns Hopkins)</i> - Optimized and SMD simulations for unfolding proteins by co-developing a variable-sized solvent box method, reducing computation time by (~20%). - Automated running SMD simulations on High Performance Computing cluster using Bash and Python scripts.	Sep 2020 – Dec 2021
Undergraduate Intern <i>Advisor : Dr. Bhushan Toley (Indian Institute of Science, Bengaluru, India)</i> - Identified reaction products and pathways involved in loop mediated and isothermal amplification (LAMP) of DNA used in PCR free diagnosis kits of tuberculosis and COVID-19. - Developed an algorithm to simulate the devised stoichiometric model and validated model outputs using experimental gel electrophoresis images and fluorescence readouts.	Jan 2019 – Dec 2019

Publications

- Thota, N. K.**; Hernandez, R. Nested Autoencoders and Nested Heteroencoders for Multi-scale Property Prediction of Perovskite Solar Cells and Photocatalysts (in preparation)
- Baird, S.; ... **Thota, N. K.**; ... Zou, Y. Bayesian Optimization Hackathon for Chemistry and Materials. *ChemRxiv*, June 23, 2025. DOI: [10.26434/chemrxiv-2023-tkcd9](https://doi.org/10.26434/chemrxiv-2023-tkcd9)
- Priyadarshini, M. S.; **Thota, N. K.**; Hernandez, R. ReLMM: Reinforcement Learning Optimizes Feature Selection in Modeling Materials. *J. Chem. Inf. Model.* **2025**, 65 (1), 153–161. DOI: [10.1021/acs.jcim.4c01934](https://doi.org/10.1021/acs.jcim.4c01934)
- Thota, N. K.**; Priyadarshini, M. S.; Hernandez, R. NestedAE: Interpretable Nested Autoencoders for Multi-Scale Materials Characterization. *Mater. Horiz.* **2024**, 11 (3), 700–707. DOI: [10.1039/D3MH01484C](https://doi.org/10.1039/D3MH01484C).
- Thota, N. K.**; Quirk, S.; Zhuang, Y.; Stover, E. R.; Lieberman, R. L.; Hernandez, R. Correlation between Chemical Denaturation and the Unfolding Energetics of Acanthamoeba Actophorin. *Biophysical Journal* **2022**, S0006349522038590. DOI: [10.1016/j.bpj.2022.11.2941](https://doi.org/10.1016/j.bpj.2022.11.2941)
- Zhuang, Y.; **Thota, N. K.**; Quirk, S.; Hernandez, R. Implementation of Telescoping Boxes in Adaptive Steered Molecular Dynamics. *J. Chem. Theory Comput.* **2022**, 18 (8), 4649–4659. DOI: [10.1021/acs.jctc.2c00498](https://doi.org/10.1021/acs.jctc.2c00498)
- Kaur, N.; **Thota, N. K.**; Toley, B. J. A Stoichiometric and Pseudo Kinetic Model of Loop Mediated Isothermal Amplification. *Comput. Struct. Biotechnol. J.* **2020**, 18, 2336–2346. DOI: [10.1016/j.csbj.2020.08.020](https://doi.org/10.1016/j.csbj.2020.08.020)

Conference Presentations

1. **Thota, N. K. (Presenter)**, Priyadarshini, M. S., Hernandez, R *Multiscale Property Prediction and Design of Perovskites Using Nested Autoencoders*. Presented at the American Institute of Chemical Engineers, Boston, MA, Nov 5, 2025 (Oral)
2. **Thota, N. K. (Presenter)**, Hernandez, R *Multi-Task Learning Using Nested Autoencoders for Designing Perovskites*. Presented at the Johns Hopkins Data Science and AI Institute Symposium, Baltimore, MD, Sep 16, 2025 (Poster)
3. **Thota, N. K. (Presenter)**, Priyadarshini, M. S., Hernandez, R *Multiscale Modeling of Materials Using Nested Autoencoders*. Presented at the AI-X Foundry Symposium, Johns Hopkins University, Baltimore, MD, Sep 27, 2023 (Poster)
4. **Thota, N. K. (Presenter)**, Quirk, S.; Zhuang, Y.; Hernandez, R. *Mutational Assay of an Actophorin Protein Using Adaptive Steered Molecular Dynamics*, Presented at American Chemical Society Conference, Chicago, IL, Aug 22, 2022 (Oral)

Teaching Experience

Instructor for Software Carpentry

Sep 2024 – Dec 2024

Johns Hopkins University, Baltimore, MD

- Taught introductory python programming and applications to 30 graduate students.
- Managed a team of a graduate TA and grader.

Instructor for Python Summer School

Jul 2023 – Aug 2023

Virtual (<https://scotch.wangyq.net>)

- Volunteered to teach an hour long session using google collab notebooks on tensor operations and backpropagation in PyTorch as part of summer school led by Dr. Pratyush Tiwary.

Teaching Assistant for Process Dynamics and Control

Sep 2022 – Dec 2022

Johns Hopkins University, Baltimore, MD

- Graded undergraduate weekly assignments and tutored 17 undergraduates during weekly recitation sessions.
- Gave lecture on applications of machine learning in process control.

Mentorship

Bryan Zhan

Jul 2024 – Sep 2024

- Bryan is a highschooler who interned at PNNL at the same time I did.
- Taught him fundamentals of coding and Python and introduced him to basics of calculus.

Awards and Certificates

1. Department graduate TA award for Software Carpentry Apr 2025
2. Among 12 finalists selected for 2022 Empower Your Pitch (EYP) competition (3 minute research communication competition). Presented "Discovery of New Solar Cell Material using Deep Learning" to university wide audience. Apr 2022

Academic Service and Leadership

Executive Board Member in Graduate Student Liason Committee (GSLC)

Sep 2022 – Sep 2023

Johns Hopkins University, Baltimore, MD, USA

- Facilitated two way communication between graduate students in the department and the university wide Graduate Research Organization (GRO).
- Organized student led seminars for graduate students in the department.

Voluntary Services

- Reviewer for AIChE Journal, JCTC and ML4LMS Workshop (Part of ICML 2024).
- Wrote project proposals for Smart Manufacturing in Merck Innovation Cup for challenges focused on digital twins and Raman Spectroscopy.

Skills

ML Models : Autoencoders, XGBoost, Gaussian Processes, **ML Interpretability** : SHAP

Optimization Methods : Reinforcement Learning, Bayesian Optimization

Coding Languages (Packages) : Python (PyTorch, TensorFlow, NumPy, SHAP), Bash, MATLAB (SimBiology, Parallel Computing), C++ , LaTeX, Markdown

Simulation Software : NAMD, **Visualization Software** : VMD, PyMOL, OVITO

Enhanced Sampling : Steered Molecular Dynamics